Machine Learning

Dr. Shylaja S S

Professor

Department of Computer Science & Engineering

Acknowledgement : Dr. Rajanikanth K (Academic Advisor, PESU)

Teaching Assistant (Arya Rajiv Chaloli)

What is Random Forest?

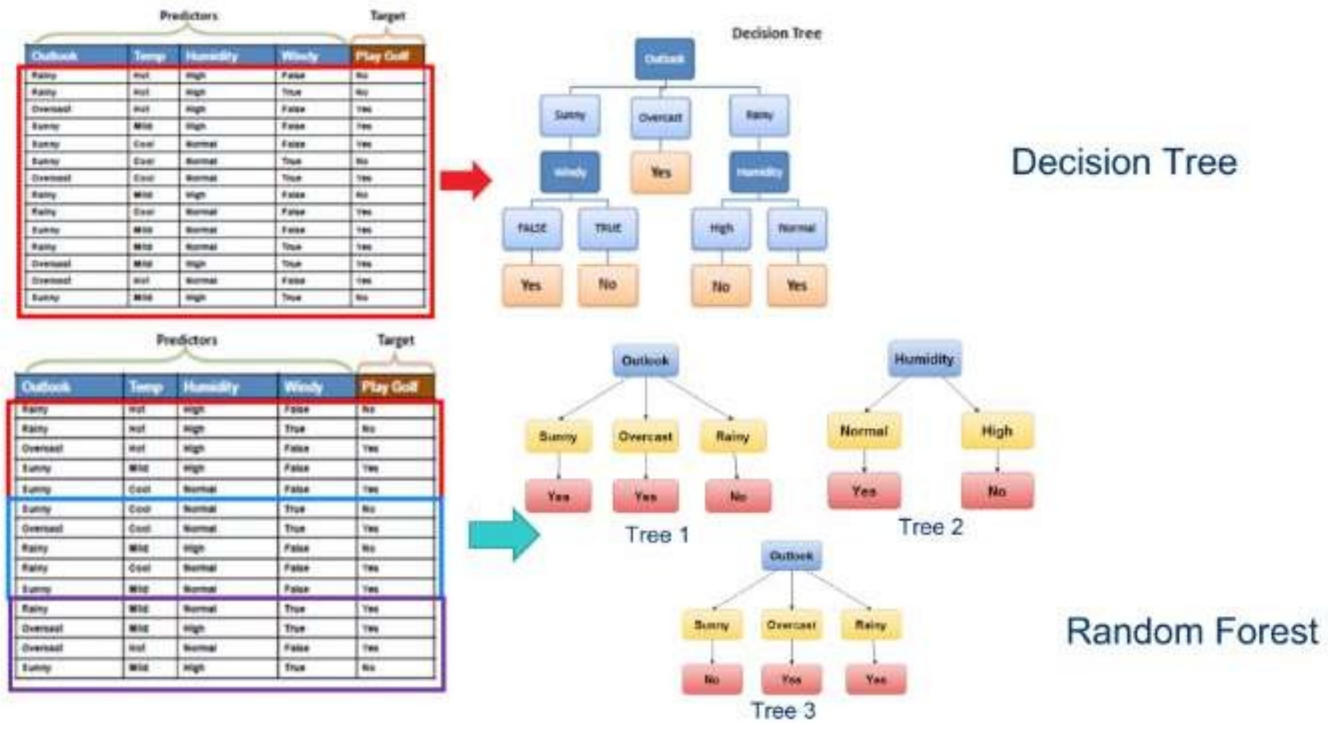
- An ensemble classifier using many decision tree models.
- Can be used for Classification or Regression.
- Accuracy and Variable importance information is provided with the results.

Ref.: https://www.google.com/

Ensemble Learning – Random Forest

Decision Tree : One Tree

Random Forest : More than one Tree



Intuition Behind Decision Tree and Random Forest:

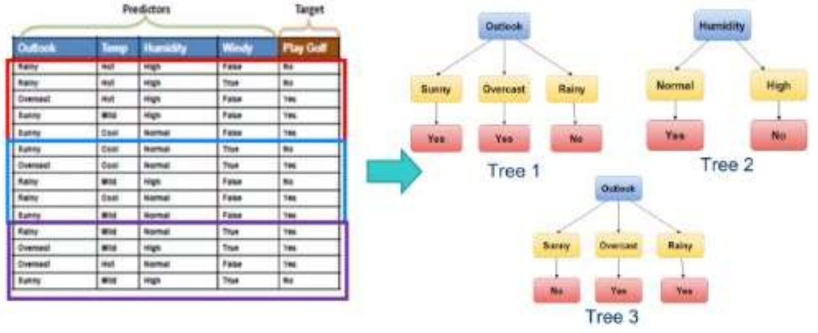
For the given test input data using Decision Tree:

Outlook	Temp.	Humidity	Windy	Play Golf
Rainy	Mild	High	False	?

Result : No

For the given test input data using Random Forest:

Outlook	Temp.	Humidity	Windy	Play Golf
Rainy	Mild	High	False	?



Tree 1 : No
Tree 2 : No
Tree 3 : Yes

Yes : 1
No : 2

Result : No

The working of Random Forest algorithm is as follows:

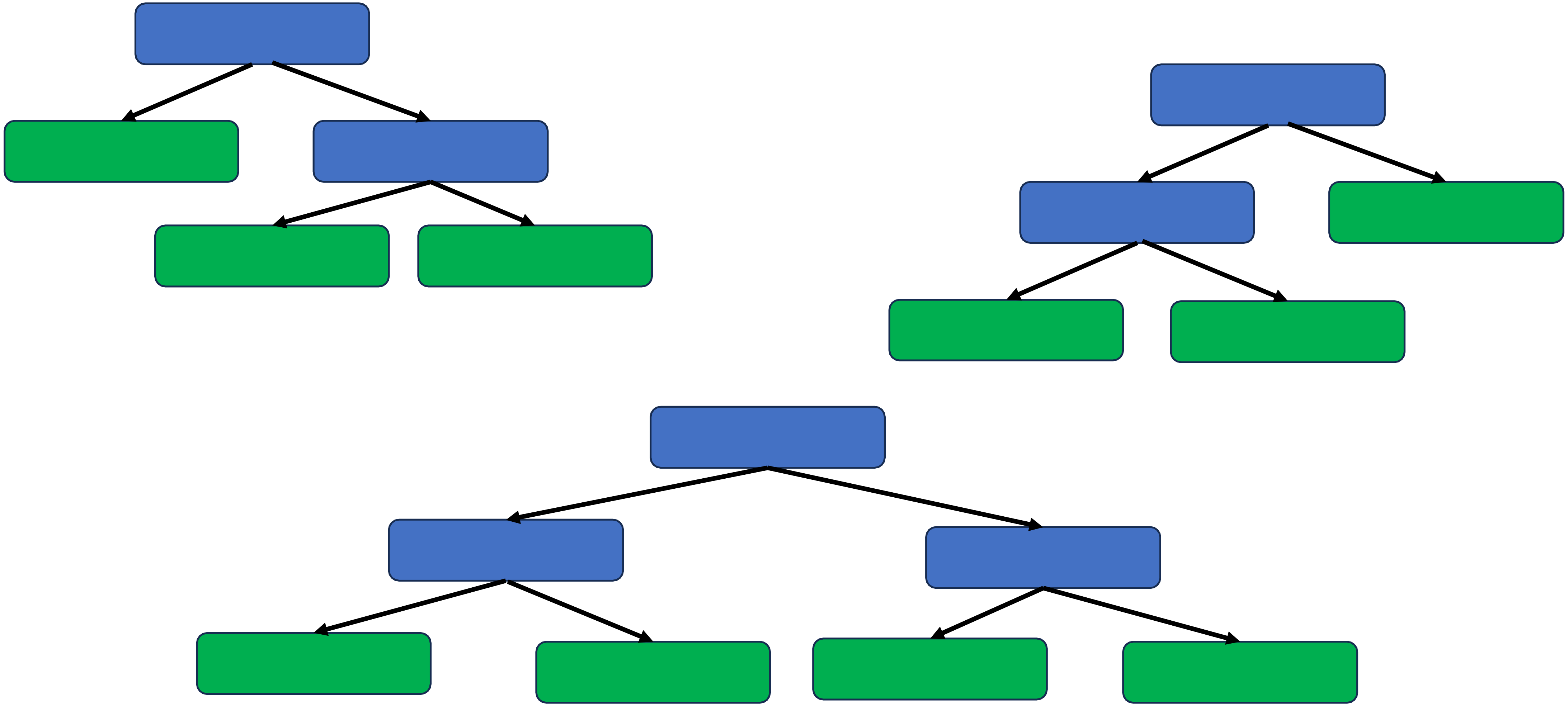
1. A random seed is chosen which pulls out at random a collection of samples from the training dataset while maintaining the the class distribution.

2. With this selected data set, a random set of attributes from the original data set is chosen based on user defined values. All the input variables are not considered because of enormous computation and high chances of over fitting.

3. In a dataset where **M is the total number of input attributes** in the dataset, **only R** attributes are chosen at random for each free where **R < M**.

4. The attributes from this set creates the best possible split to develop a decision tree model. The process repeats for each of the branches until the termination condition stating that leaves are the nodes that are too small to split.

Ensemble Learning – Random Forest



Ensemble Learning – Random Forest

1. Create Bootstrapped dataset
2. Create decision trees using a random subset of features
3. Compute the out of Bag Error
4. Run the Samples through the Random Forest and classify based on Voting

- Remember, in bootstrapping we sample with replacement, and therefore not all observations are used for each bootstrap sample. On average 1/3 of them are not used!

- We call them out-of-bag samples (OOB)

- We can predict the response for the i-th observation using each of the trees in which that observation was OOB and do this for n observations

- Calculate overall OOB MSE or classification error

Original Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	Yes	167	Yes

Original Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	Yes	167	Yes

Bootstrapped Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	Yes	Yes	180	Yes

Original Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	Yes	167	Yes

Bootstrapped Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	Yes	Yes	180	Yes
No	No	No	125	No
Yes	No	Yes	167	Yes

Original Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	Yes	167	Yes

Bootstrapped Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	Yes	Yes	180	Yes
No	No	No	125	No

Original Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	Yes	167	Yes

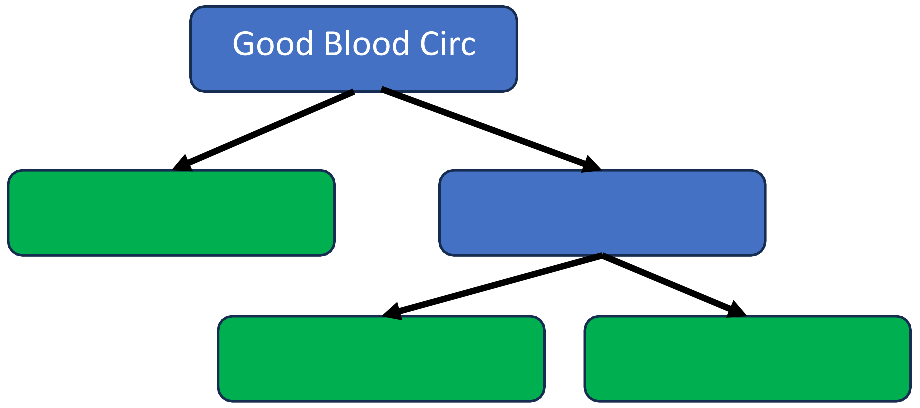
Bootstrapped Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	Yes	Yes	180	Yes
No	No	No	125	No
Yes	No	Yes	167	Yes
Yes	No	Yes	167	Yes

Bootstrapped Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	Yes	Yes	180	Yes
No	No	No	125	No
Yes	No	Yes	167	Yes
Yes	No	Yes	167	Yes

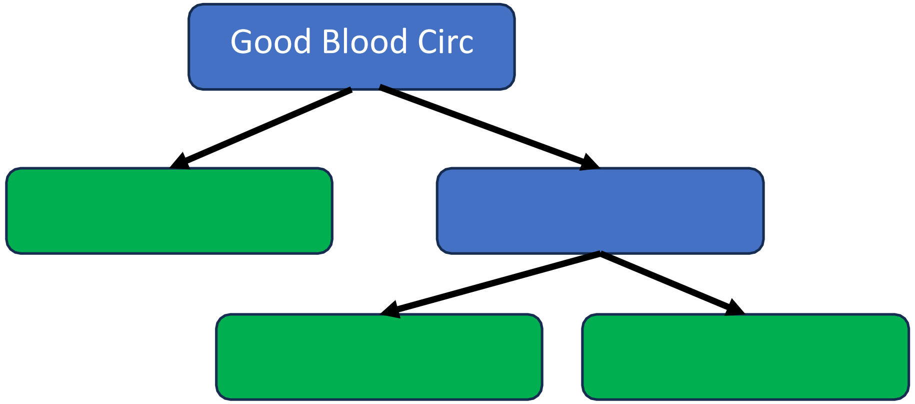
Random subset of features

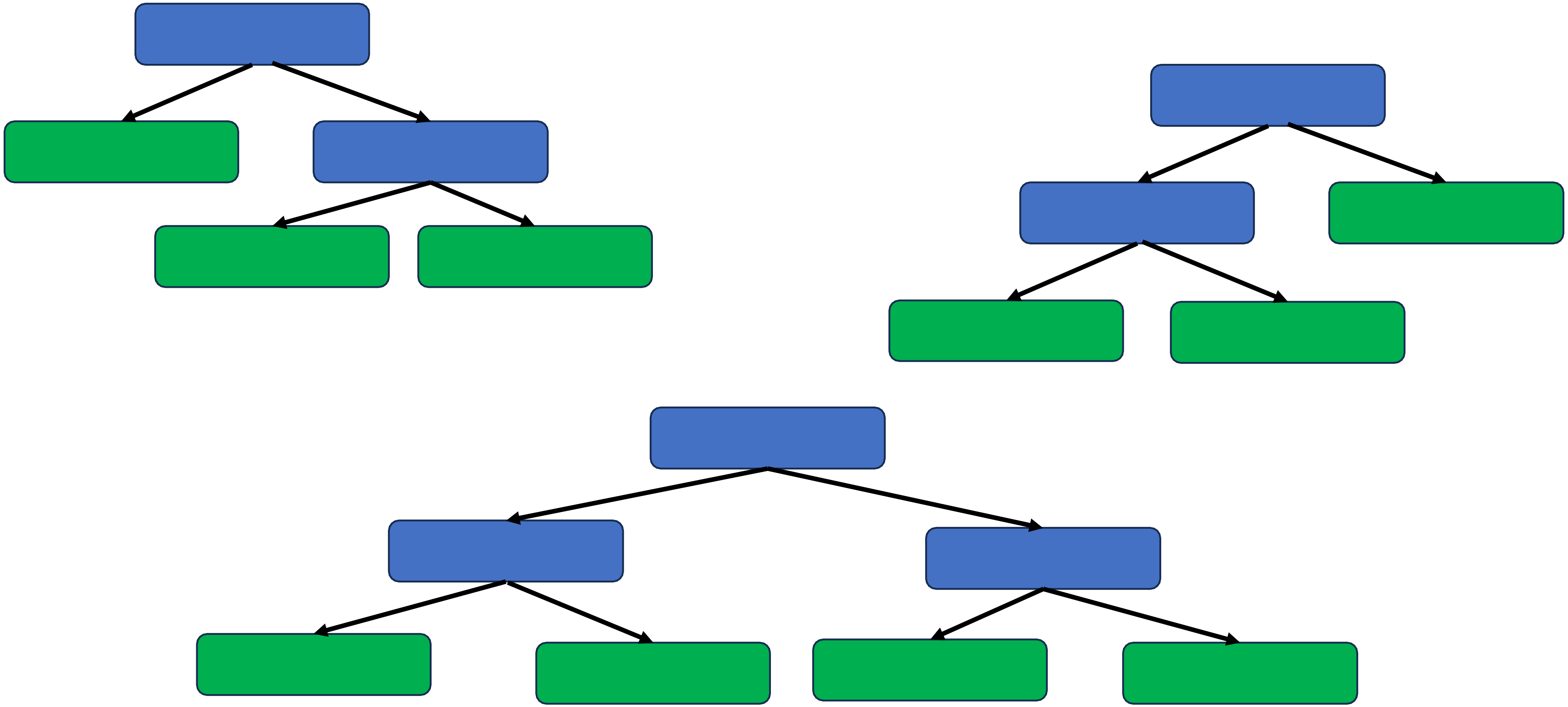


Bootstrapped Dataset

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	Yes	Yes	180	Yes
No	No	No	125	No
Yes	No	Yes	167	Yes
Yes	No	Yes	167	Yes

Random subset of features





Advantages of Random Forest:

- No need for pruning trees
- Accuracy and variable importance generated automatically
- Overfitting is not a problem
- Not very sensitive to outliers in training data
- Easy to set parameters
- Good performance

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	???	???	No

Missing Data

1. Missing Data in the training sample.
2. Missing data in the test sample

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	???	???	No

Did not have heart disease

No is the most common value for Blocked Arteries

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	NO	???	No

Did not have heart disease

No is the most common value for Blocked Arteries

Initial Guess

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	NO	167.5	No

Did not have heart disease

Weight is numeric

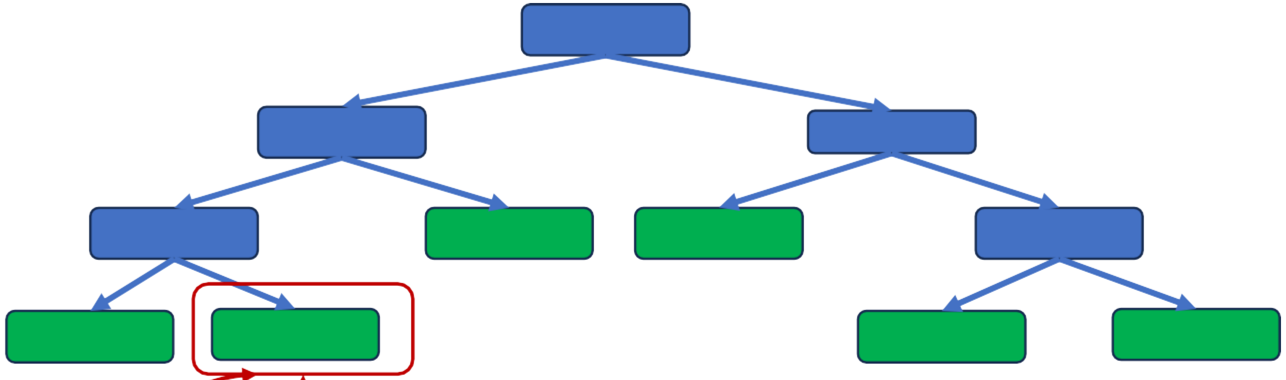
Initial Guess (Median)

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	No	167.5	No

Determine which samples are similar to the one with missing data

Ensemble Learning – Random Forest

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	No	167.5	No

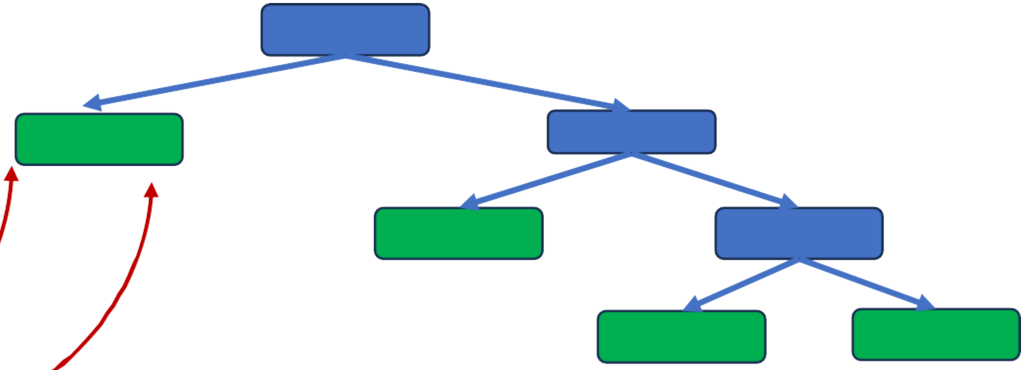


Proximity Matrix

	1	2	3	4
1				
2				
3				1
4			1	

Ensemble Learning – Random Forest

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	No	167.5	No



Proximity Matrix

	1	2	3	4
1				
2			1	1
3		1		2
4		1	2	

	1	2	3	4
1		2	1	1
2	2		1	1
3	1	1		8
4	1	1	8	

	1	2	3	4
1		**0.2**	**0.1**	**0.1**
2	**0.2**		**0.1**	**0.1**
3	**0.1**	**0.1**		**0.8**
4	**0.1**	**0.1**	**0.8**	

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	NO	167.5	No

Compute Weighted frequency

Yes: 1/3
No: 2/3

Weight of Yes: (proximity of yes) / all proximities
= 0.1/1=0.1
Weight of No: (proximity of no) / all proximities
= 0.9/1 = 0.9

Weighted frequency of yes : 1/3 * weight of yes
= (1/3) *0.1 = 0.03
Weighted frequency of no : 2/3 * weight of no
= (2/3) *0.9 = 0.6

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	Yes	NO	198.5	No

Compute Weighted frequency

	1	2	3	4
1		0.2	0.1	0.1
2	0.2		0.1	0.1
3	0.1	0.1		0.8
4	0.1	0.1	0.8	

Weighted average : (0.1 * 125) + (0.1 * 180) + (0.8 *210)
= 198.5

New Sample

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	No	???	168	

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	No	???	168	Yes

Chest Pain	Good Blood Circ	Blocked Arteries	Weight	Heart Disease
Yes	No	???	168	No

Ensemble Learning – Random Forest – Example

Dataset

Predict whether a student Passes based on Hours Studied and Sleep Hours.

Hours Studied	Sleep Hours	Pass
1	5	No
2	6	No
3	7	Yes
4	6	Yes
5	5	Yes
6	4	Yes

Ensemble Learning – Random Forest – Example

Step 1. Build Decision Trees on Bootstrapped Data + Random Features

Tree 1 (trained on random sample + only "Hours Studied"):

If `Hours Studied ≥ 3` → Yes, else No.

Tree 2 (trained on random sample + only "Sleep Hours"):

If `Sleep Hours ≥ 6` → Yes, else No.

Tree 3 (trained on random sample + both features):

If `Hours Studied ≥ 4` → Yes, else No.

Hours Studied	Sleep Hours	Pass
1	5	No
2	6	No
3	7	Yes
4	6	Yes
5	5	Yes
6	4	Yes

Step 2. Each Tree Votes

Suppose we predict for: Student with 3 hours studied, 6 hours sleep

Tree 1 → Yes (≥3 hrs studied)
Tree 2 → Yes (≥6 hrs sleep)
Tree 3 → No (<4 hrs studied)

Step 3. Final Prediction by Majority Vote

Votes: Yes, Yes, No → Final = **Yes** (**Pass**)

Hours Studied	Sleep Hours	Pass
1	5	No
2	6	No
3	7	Yes
4	6	Yes
5	5	Yes
6	4	Yes
3	**6**	**?**

Machine Learning

Dr. Shylaja S S

Department of Computer Science & Engineering

shylaja.sharath@pes.edu